

Reducing Customer Churn at Company A

A profit-driven, calibrated, and fair churn model.

Telecom Retention Analytics

100,000 customers | 123 engineered behavioral features | demographics excluded by design

EXECUTIVE SUMMARY

The bottom line

REALISTIC MODEL

AUC 0.70

Tuned + calibrated XGBoost. No leakage, no 0.99 fantasy.

TARGETING LIFT

1.34x

vs. random outreach, ranking by expected value at risk.

BASE-CASE ROI

2.4x

Net \$189,190 at the profit-optimal 42% contact depth.

What changed from a naive first pass:

Protected attributes removed, probabilities calibrated, scores re-weighted to the true ~2% churn rate, the decision threshold set on a profit curve, drivers explained with SHAP, and the business case shown as bear / base / bull with cited assumptions.

PROBLEM FRAMING

Churn is expensive - but not every customer is worth the same call

The task.

Predict which subscribers will churn, explain why, and turn that into a retention plan that makes money.

The data.

100,000 customers, 100 raw columns across account, usage, and demographic tables.

The honesty problem.

The sample is balanced to ~50% churn for training, but real monthly churn is ~2%. We train on the sample, then re-weight predictions back to reality so the dollars are credible.

KEY NUMBERS

ARPU (avg monthly revenue)	\$59
Median tenure	16 months
Model features (engineered)	123
Protected attributes dropped	20
Assumed true monthly churn	2%

RESPONSIBLE MODELING

We deliberately excluded demographics

Targeting retention by ethnicity, income, marital status, or dwelling is a legal and ethical hazard - and it isn't actionable. All 20 such attributes are dropped before the model sees the data, with an automated guard that fails the build if one leaks through encoding.

EXCLUDED

- ethnic
- crclscod
- kid11_15
- dwlltype
- infobase

EXCLUDED

- marital
- kid0_2
- kid16_17
- dwllsize
- numbcars

EXCLUDED

- income
- kid3_5
- adults
- lor
- truck

EXCLUDED

- creditcd
- kid6_10
- ownrent
- HHstatin
- rv

APPROACH

One reproducible pipeline, end to end

1 Engineer

Merge usage + account tables; build behavioral features (equipment age, usage trend, care intensity, overage share).

2 Model

Compare Logistic Regression, Random Forest, XGBoost; tune XGBoost with randomized search + 5-fold CV.

3 Calibrate

Isotonic calibration so scores are real probabilities; re-weight to the true 2% base rate.

4 Decide

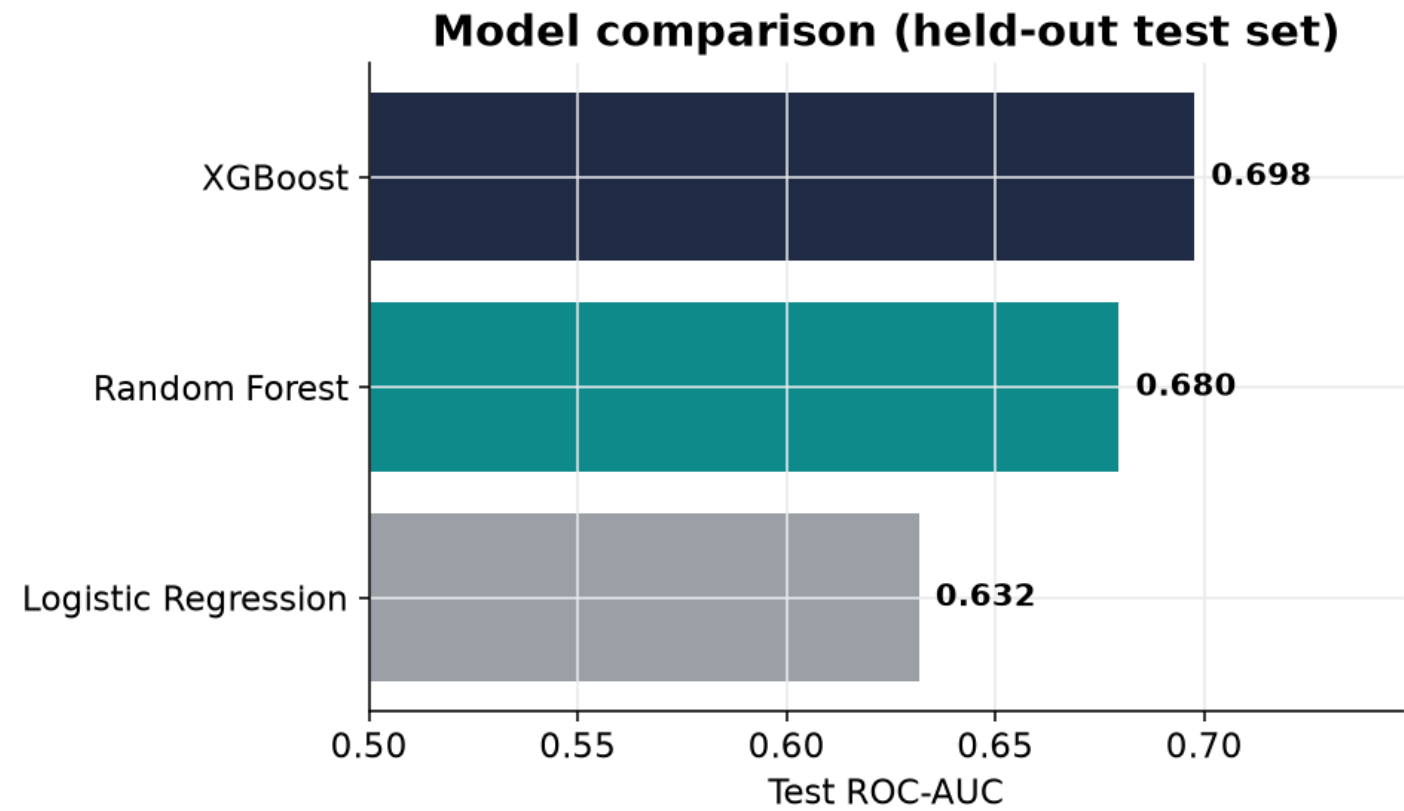
Rank by expected value at risk; choose contact depth on the profit curve, not an arbitrary 0.5 cutoff.

5 Explain & act

SHAP drivers, K-means personas, and an uplift/persuadables view feed a bear/base/bull business case.

MODEL SELECTION

XGBoost wins - at a believable accuracy



Why this is credible

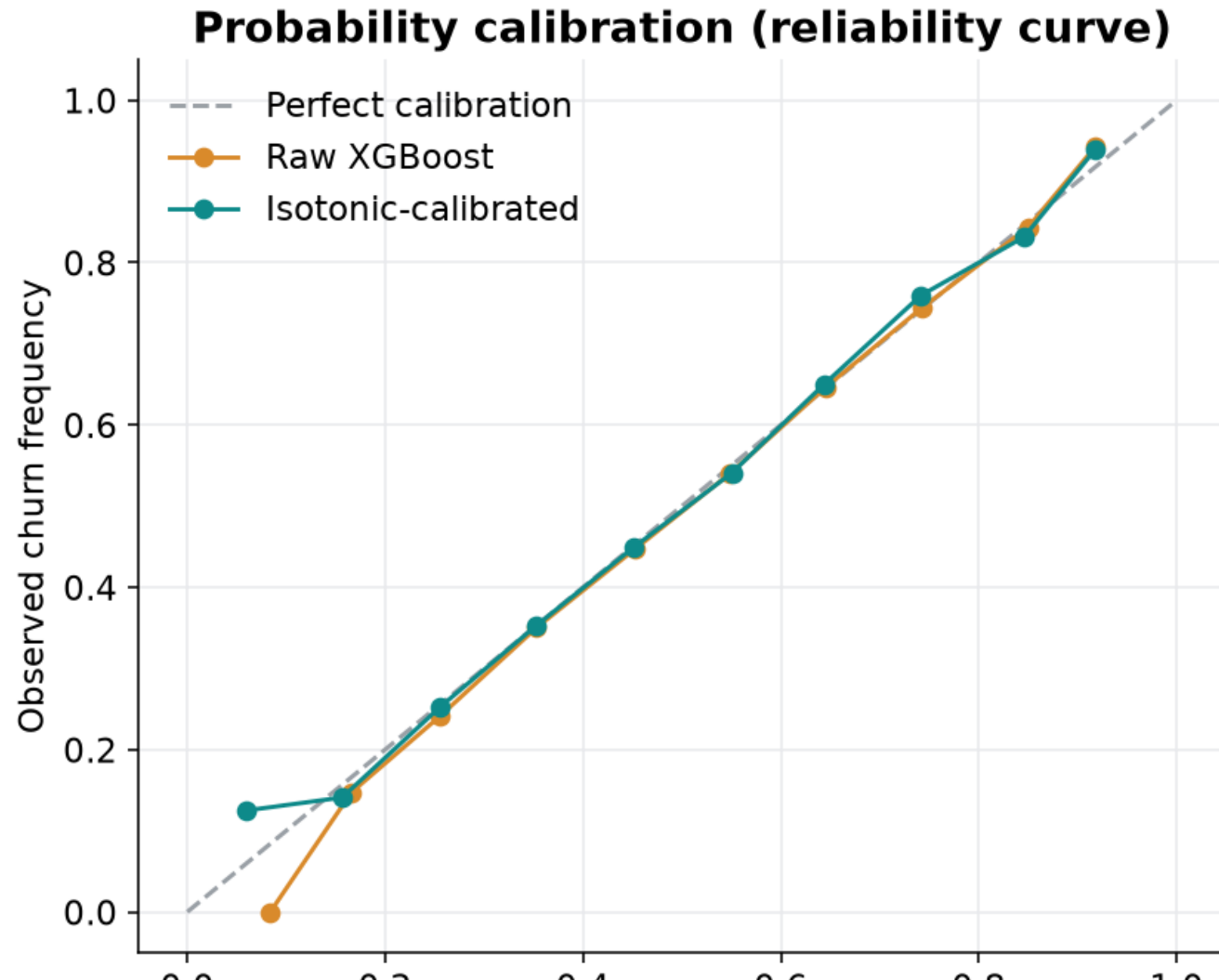
AUC 0.70, PR-AUC 0.69. An AUC near 0.70 is normal for telecom churn on behavioral data - a 0.95+ score would signal target leakage.

5-fold CV: 0.699 +/- 0.003. Tiny variance across folds means the model is stable, not lucky.

Brier score improves after calibration - the probabilities can be trusted for dollar math.

TRUSTWORTHY PROBABILITIES

Calibration + the base-rate correction



A score is not a probability

Raw model scores cluster near 0.5 because the training sample is balanced. Isotonic calibration aligns predicted probability with observed frequency (the diagonal).

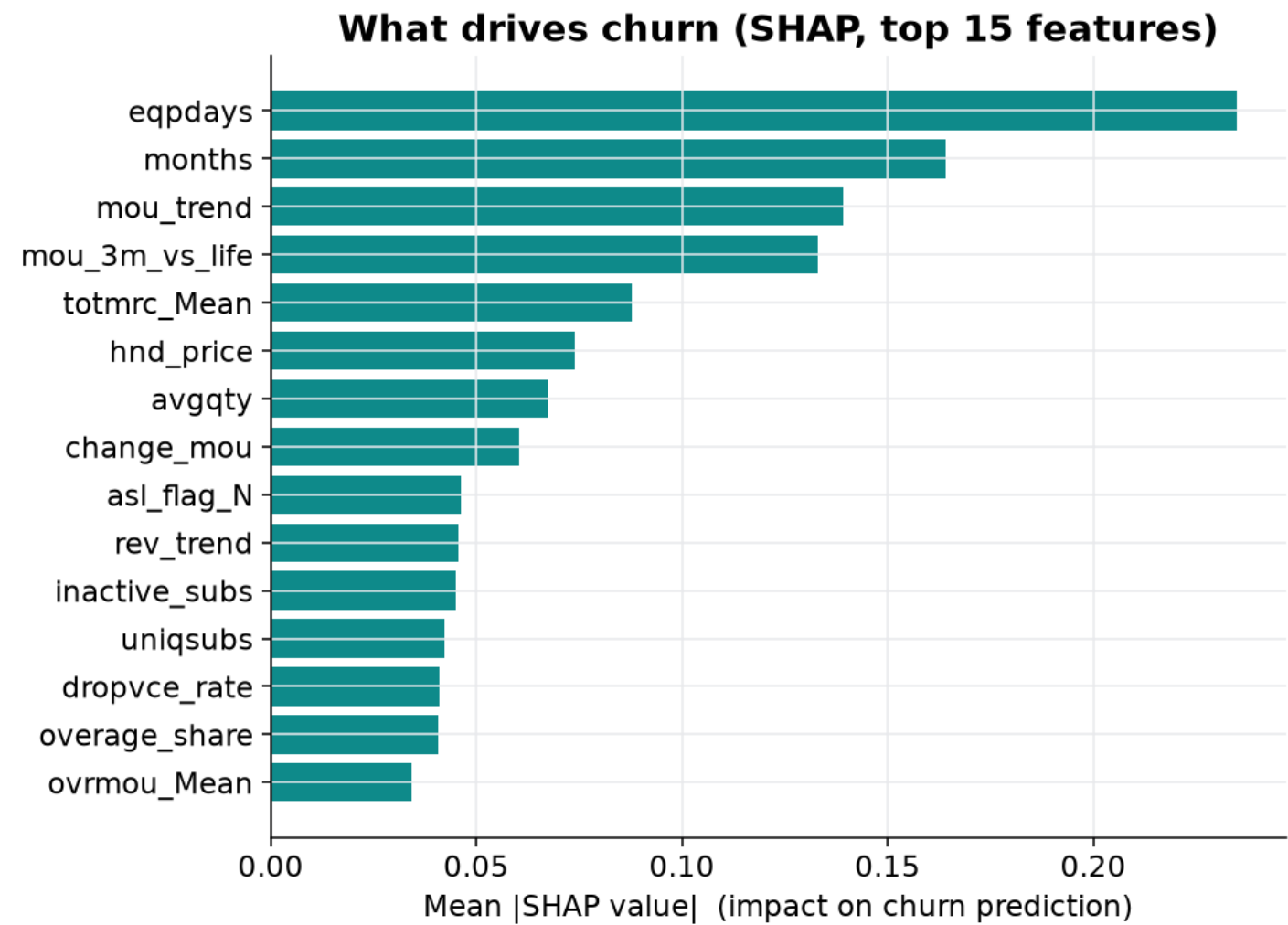
Then we correct the prior:

mean $P(\text{churn})$ 0.50 in the sample $\rightarrow$ 0.027 re-weighted to the real ~2% rate.

Without this step, every revenue-at-risk and ROI figure would be inflated by ~20x.

WHY CUSTOMERS LEAVE

SHAP - the model's honest drivers



All actionable, none demographic

Equipment age (eqpdays)

- the top driver. Old handsets churn; proactive upgrade offers are the lever.

Tenure & usage trend

- declining minutes-of-use flags disengagement early.

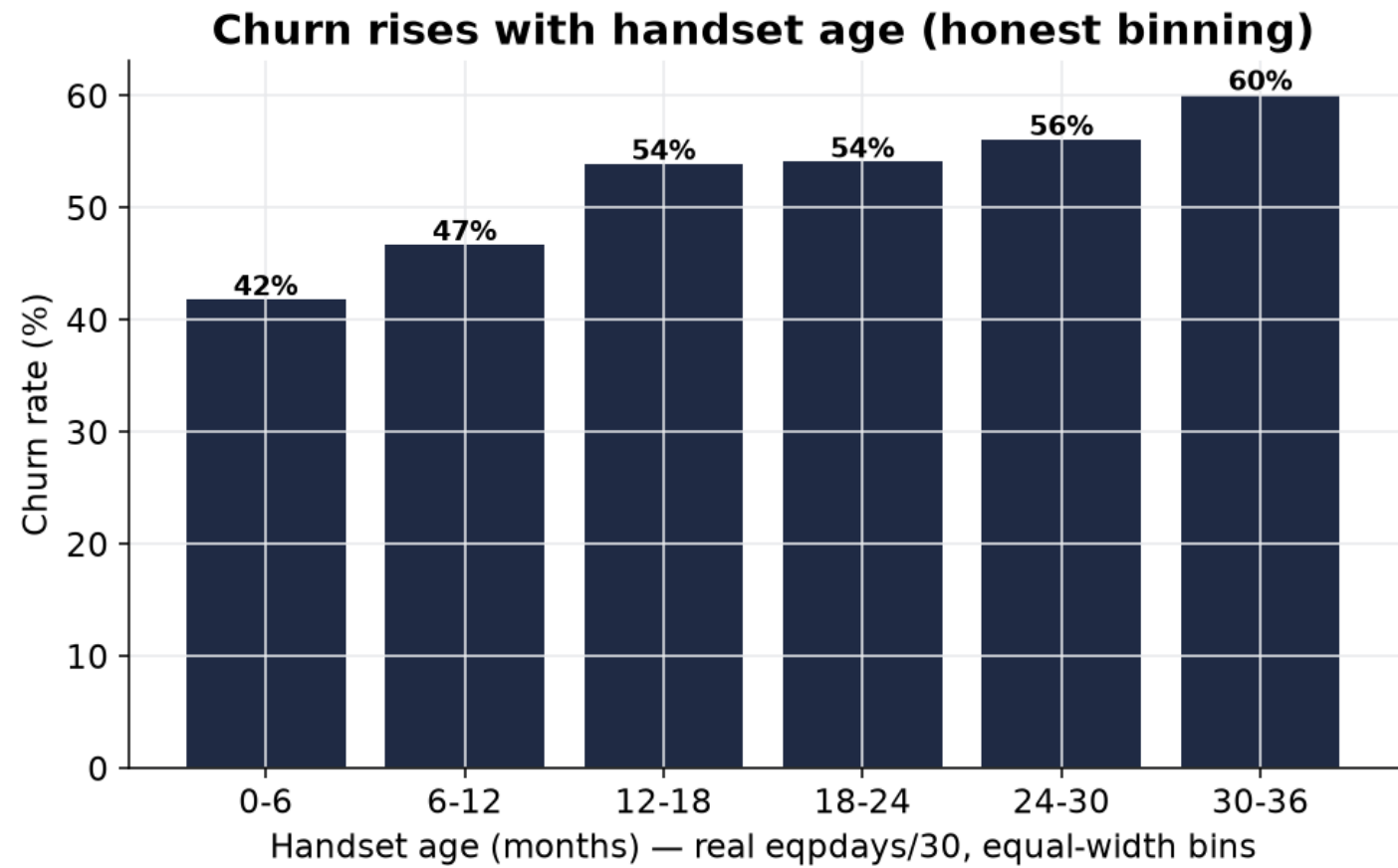
Care intensity & dropped calls

- service quality and support friction. Fixable operationally.

Because demographics were excluded, every top driver maps to an action the business can actually take.

HONEST SEGMENTATION

Churn rises with equipment age



Binned, not over-precise

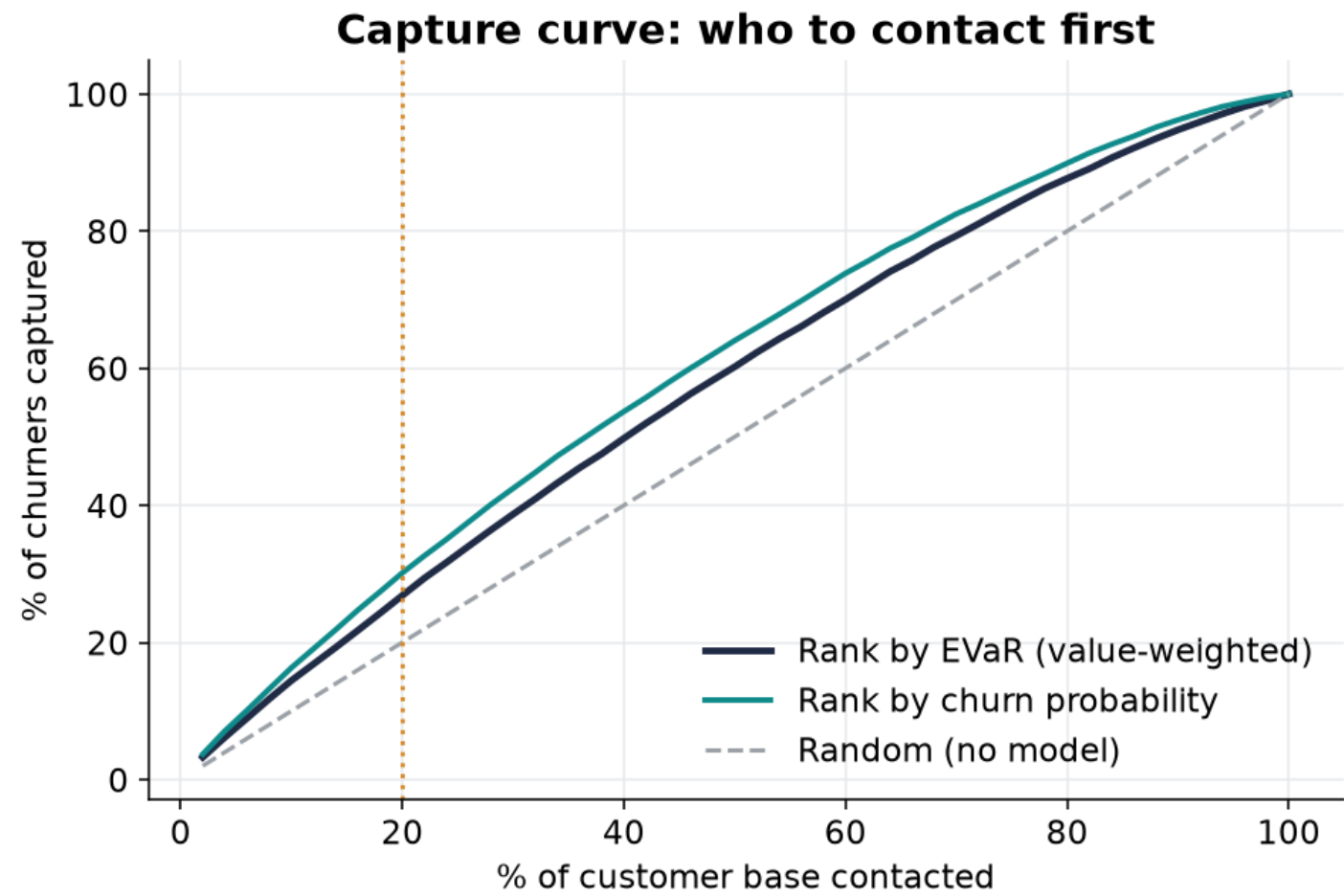
Equipment age is grouped into interpretable buckets rather than implying false day-level precision.

The pattern is monotonic and clean: the longer a customer keeps an aging device, the more likely they leave.

Action: trigger an upgrade offer as devices cross the high-risk age band.

PRIORITIZE

Call the highest-value risk first



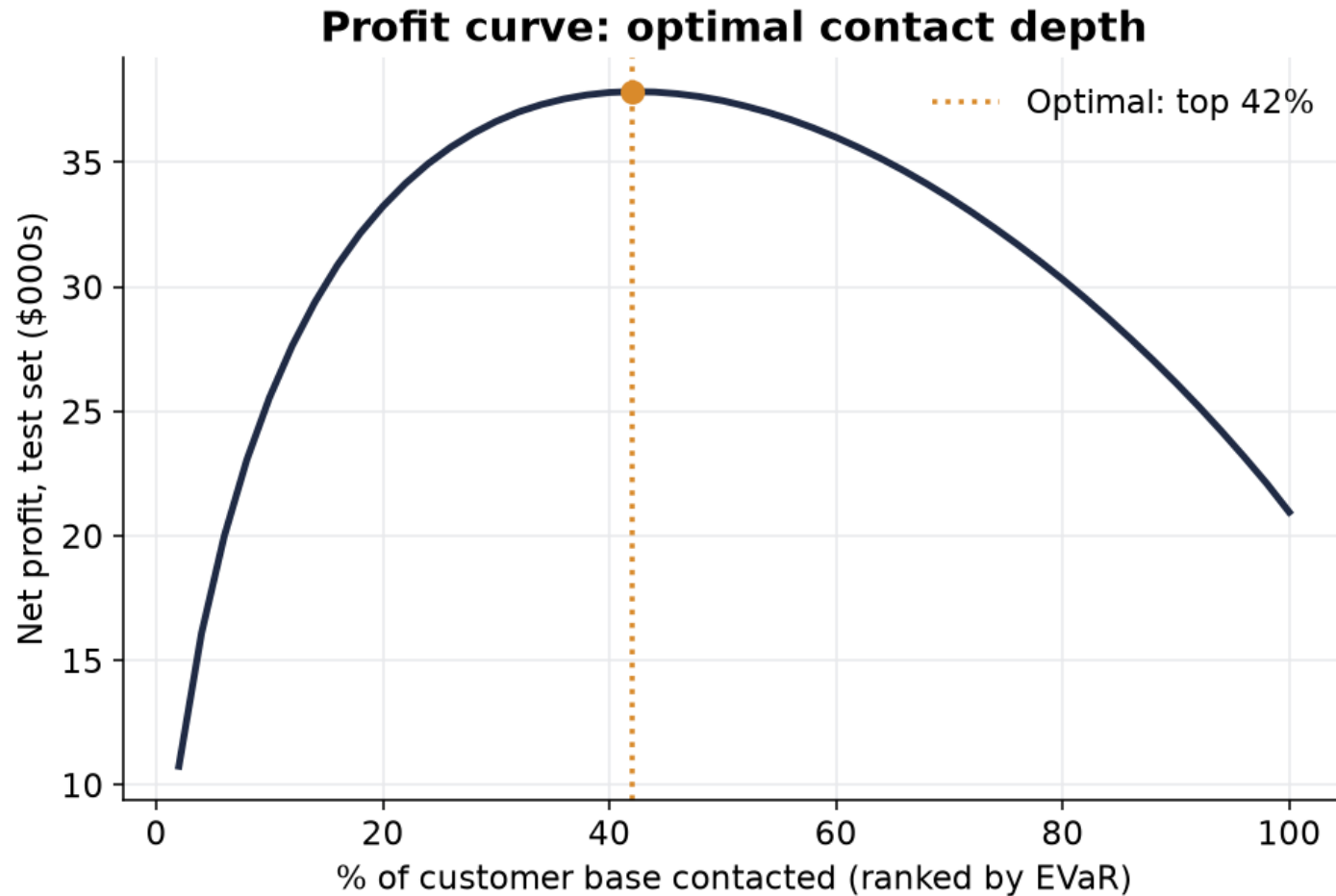
Expected value at risk

We rank by $P(\text{churn}) \times \text{margin} \times \text{lifetime value}$, not probability alone - a \$100 customer at 30% risk beats a \$20 customer at 60%.

Contacting the top 20% by value captures 27% of churn value - 1.34x better than random.

OPTIMIZE

How deep to go - the profit curve



Maximize profit, not recall

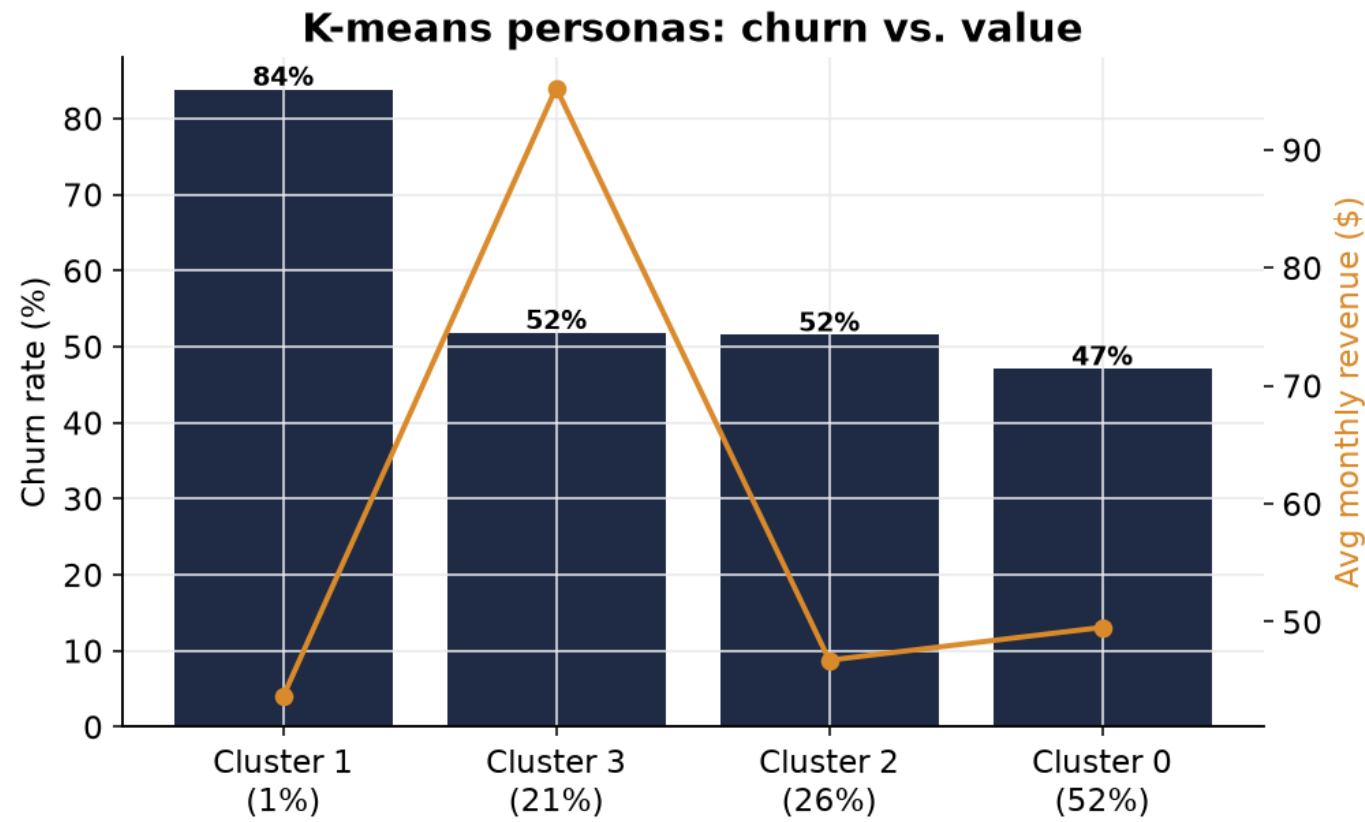
Calling everyone wastes incentives; calling too few leaves value on the table. We sweep contact depth and pick the peak.

Optimal depth: 42%

Net profit \$189,190 - ROI 2.4x - ~351 customers saved (base case).

WHO THEY ARE

Four behavioral personas (K-means)



Behavioral, not demographic

Clusters are built only from behavior - usage trend, equipment age, care intensity, overage, tenure.

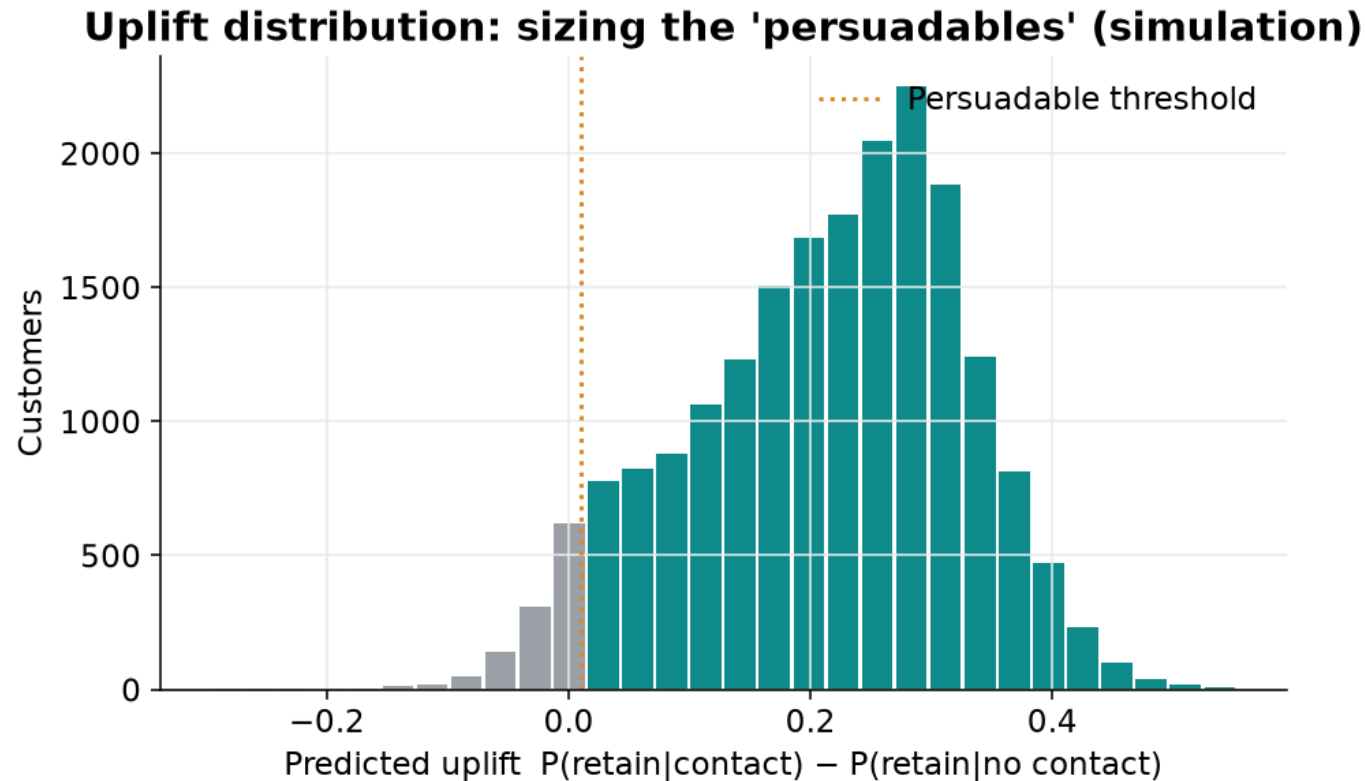
High-risk segment

churn rate 84%, high care-call intensity and aging equipment - the clearest save target.

A separate high-ARPU segment justifies premium, concierge-style retention offers.

PERSUADABLES

Treat customers a campaign can actually move



Beyond risk: responsiveness

Some high-risk customers leave regardless; the money is in persuadables - those whose retention an offer actually changes (T-learner uplift).

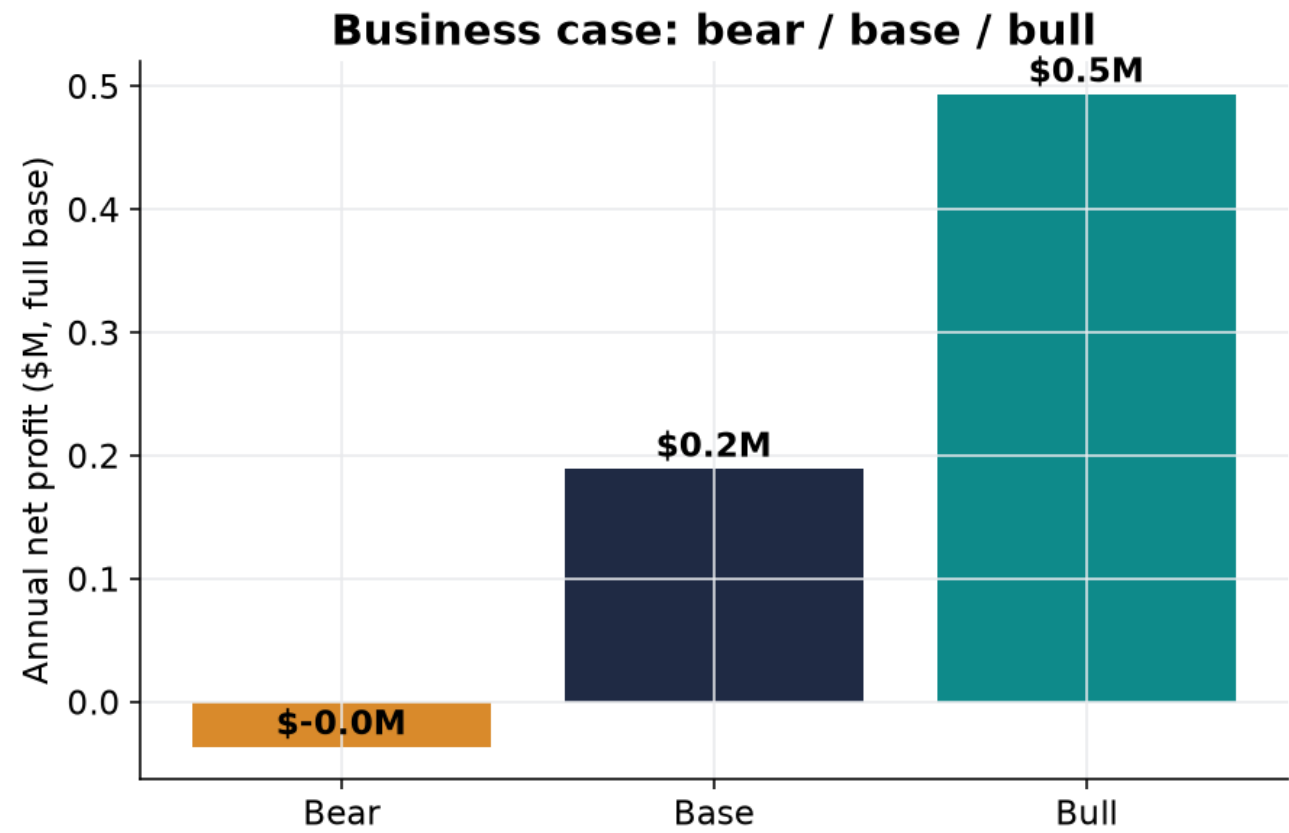
Honesty note

This data is observational (no test/control), so the response here is a clearly-labeled simulation to demonstrate the method.

Next step: a live A/B holdout to measure real uplift before scaling.

THE MONEY

Bear / base / bull - profit-positive in the base & bull



Assumptions are cited, not invented

Each scenario varies save rate, offer cost, margin, and CLV horizon - all sourced in REFERENCES.md.

Bear	\$-36,379
save 12% - \$35 offer	ROI 0.8x - ~211 saved
Base	\$189,190
save 20% - \$30 offer	ROI 2.4x - ~351 saved
Bull	\$493,383
save 28% - \$25 offer	ROI 6.1x - ~492 saved

Operationalize the model - and prove it with a test

1

Deploy value-ranked scoring

Score the base monthly; route the profit-optimal top 42% by value-at-risk to retention.

2

Trigger device-age offers

Automate upgrade offers as handsets cross the high-risk age band - the #1 SHAP driver.

3

Fix service friction

Feed dropped-call and care-intensity signals to network and support ops as leading indicators.

4

Run an A/B retention test

Randomized treatment/control to measure REAL uplift and replace the simulated persuadables figures.

5

Re-validate assumptions

Confirm save rate, offer cost, margin, and CLV with finance; refresh the bear/base/bull case quarterly.